

MODULE 01

Software Modernization Overview

Day 1 · Core MVP Module



ADOBE STOCK PLACEHOLDER

Suggested search: "legacy code on computer screen close up (wide hero banner)"

LEARNING OBJECTIVES

- Explain modernization outcomes in business and technical terms.
- Distinguish rehost, refactor, rearchitect, and rebuild.
- Identify modernization candidates in a legacy .NET application.



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Suggested search: "software architecture diagram on whiteboard (supporting image)"

WHY THIS MODULE MATTERS

This module establishes the course baseline and shared vocabulary.

Learners review a real .NET application and classify modernization opportunities by strategy and risk.

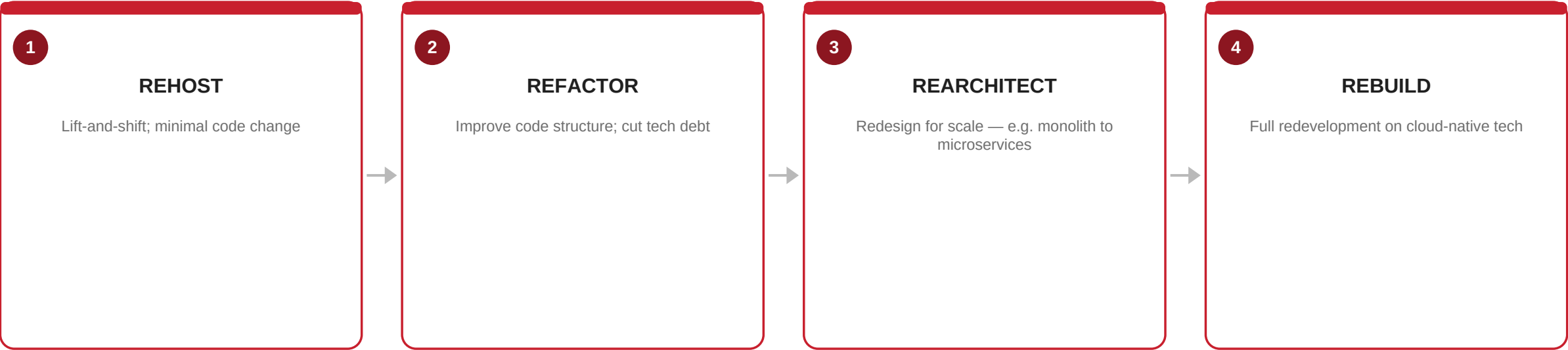


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Suggested search: "developer reviewing old codebase (concept image)"

CONCEPT MAP

Microsoft's Cloud Adoption Framework: four of the core migration strategies, in order of increasing change.



FROM THE DOCS: THE FULL MIGRATION FRAMEWORK

- Microsoft's Cloud Adoption Framework actually defines 8 strategies: Retire, Retain, Rehost, Replatform, Refactor, Rearchitect, Rebuild, Replace.
- Rehost = like-for-like migration, e.g. on-prem servers moved directly to Azure VMs.
- Microsoft's own warning: don't rehost a problematic workload as-is — it just carries the technical debt into the cloud.
- Rearchitect example Microsoft gives: breaking a monolith into microservices for scale and agility.
- Rebuild is reserved for systems too outdated or inflexible to modernize any other way.



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Suggested search: "modern vs legacy technology concept (detail image)"

COURSEWARE FOCUS

- Why modernization matters
- Rehost, refactor, rearchitect, rebuild
- How the reference app anchors the course



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THE REFERENCE APP, IN ITS OWN WORDS

- eShopOnWeb describes itself as a “sample ASP.NET Core reference application... demonstrating a single-process (monolithic) application architecture.”
- Its own README points to eShopOnContainers/eShop as the sibling app that “focuses on a microservices/containers-based application architecture.”
- Same product line, two strategies: eShopOnWeb is the rehost/refactor target; eShop is what a rearchitect looks like.



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Suggested search: “software architecture diagram on whiteboard (reference screenshot)”

TOOLS & RESOURCES

Reference Code

dotnet-architecture/eShopOnWeb

Baseline monolithic application to inspect for modernization candidates.

Documentation

Microsoft Azure Architecture Center: .NET Application Architecture Guides

Slide material explaining Rehost, Refactor, Rearchitect, and Rebuild strategies.

AI Tooling

GitHub Copilot Upgrade Agent / Extension

Used to assess the legacy solution and generate modernization plans.

SUPPORTING ASSETS

- Reference app: [course/repos/eShopOnWeb](https://github.com/ALDOT/eShopOnWeb)
- Design anchor: [course/design.md](#)



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Suggested search: "software architecture diagram on whiteboard (reference screenshot)"

LAB 01: MODERNIZATION DISCOVERY



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Suggested search: "developer reviewing old codebase (lab / hands-on photo)"

LAB STEPS

Goal: Identify and prioritize modernization candidates in the baseline app.

Tools: Git, .NET SDK, VS Code · No cloud dependency required.

- Step 1 — Open eShopOnWeb and run a local build.
- Step 2 — Identify major architecture boundaries and high-risk areas.
- Step 3 — Record at least 8 modernization candidates.
- Step 4 — Tag each item as rehost, refactor, rearchitect, or rebuild.
- Step 5 — Prioritize the list by impact and implementation effort.
- Validation: baseline app builds successfully, and the candidate matrix contains classification plus priority.



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Suggested search: "modern vs legacy technology concept (hands-on lab photo)"

KNOWLEDGE CHECK: Software Modernization Overview

Which modernization strategy moves an application to new infrastructure with minimal changes to the underlying code?

- A. Rebuild
- B. Rehost
- C. Refactor
- D. Rearchitect



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KNOWLEDGE CHECK: Software Modernization Overview

According to Microsoft's Cloud Adoption Framework, how many total migration strategies ("R's") does it define?

- A. 4
- B. 5
- C. 6
- D. 8



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SOURCES & FURTHER READING

EXTERNAL DOCUMENTATION

■ Select your cloud migration strategies — Cloud Adoption Framework

<https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/plan/select-cloud-migration-strategy>

REFERENCED IN THIS REPOSITORY

■ `course/repos/eShopOnWeb/README.md`

■ `course/design.md`

■ `course/mvp-delivery/modules/module-01-modernization-overview.md`

RECAP & SUCCESS CRITERIA

- Learner completes candidate matrix with at least 8 candidate changes and strategy labels.
- Tier: Core MVP Module



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Suggested search: "developer reviewing old codebase (team success photo)"